

2026-05-07-p2-f27-aider-voice-repomap-design

Phase 2 / Feature 27 — Aider Voice Input + Repo-Map

Date: 2026-05-07 **Status:** Approved **Programme:** CLI-Agent Fusion — Phase 2 port (aider: voice input + repo-map)

Programme position: F27 is the **seventh** Phase 2 feature (F21-F26 shipped).

1. Goal

Ship real voice input (speech-to-text) and tree-sitter-based repository mapping for the HelixCode CLI agent. Voice: arecord/sox capture → OpenAI Whisper API (with local whisper.cpp fallback). Repo-map: tree-sitter AST parsing for function/class/import declarations, cached per-commit SHA. Four tools: voice_start, voice_stop, voice_transcribe, repomap. Slash: /aider (voice/repomap subcommands).

2. Architecture

2.1 Voice Pipeline

```
Microphone → arecord/sox → temp WAV → Whisper API → text
                                     ↓ (fallback)
                                     whisper.cpp → text
```

- `internal/voice/` (NEW) — VoiceRecorder (shell capture), VoiceTranscriber (Whisper API + local fallback)
- Audio format: 16kHz mono WAV via `arecord` (Linux) or `sox` (cross-platform)
- Temp files at `$XDG_RUNTIME_DIR/helixcode/voice/` (auto-cleaned)
- Transcription: OpenAI Whisper API (env: `OPENAI_API_KEY`) → `local whisper.cpp` CLI fallback

2.2 Repo-Map Pipeline

Project dir → glob files → tree-sitter parse → extract defs → JSON map →

- `internal/repomap/` (NEW — extends existing `internal/repomap/ stub`)
- AST parsing via `go-tree-sitter` (already indirect dep in `go.sum`)
- Map includes: functions, classes, imports, exports per file
- Cache keyed by `git rev-parse HEAD`, invalidated on commit
- Output format: structured JSON map

3. Components

File	Purpose
internal/voice/types.go	VoiceRecorder, VoiceTranscriber, config, sentinels
internal/voice/recorder.go	Shell-based audio capture (arecord/sox)
internal/voice/transcriber.go	Whisper API + whisper.cpp fallback
internal/voice/voice_tools.go	voice_start/stop/transcribe tools
internal/repomap/mapper.go	Tree-sitter AST parser + map builder
internal/repomap/cache.go	Per-commit map cache
internal/repomap/repomap_tool.go	repomap tool implementation
internal/commands/aider_command.go	/aider slash command

4. Anti-Bluff Hot Zone

1. **Voice recorded but 0 bytes** — WAV file size > 44 bytes (header minimum)
 2. **Transcription returns empty** — text output is non-empty
 3. **Whisper API fallback silent** — fallback path is exercised when API key absent
 4. **Repo-map returns empty for non-empty repo** — at least 1 function/class discovered
 5. **Cache returns stale data after commit** — map differs pre/post commit
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5. Task Breakdown

#	Task	Description
T01	Bootstrap	F27 evidence + advance PROGRESS + CONTINUATION
T02	voice/types.go + recorder.go	VoiceRecorder with arecord/sox (TDD)
T03	voice/transcriber.go	Whisper API + whisper.cpp fallback (TDD)
T04	voice/voice_tools.go	voice_start/stop/transcribe tools (TDD)
T05	repomap/mapper.go + cache.go	Tree-sitter AST map + commit cache (TDD)
T06	repomap/repomap_tool.go	repomap tool (TDD)
T07	/aider slash	voice/repomap subcommands (TDD)

#	Task	Description
T08	main.go wiring	Register tools + slash
T09	Challenge harness + close-out	6 phases + push 4 remotes

F27 spec — Aider Voice Input + Repo-Map.